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The Vertex AI Readiness Guide

Is your organisation ready to build AI agents on Google Cloud? This guide covers the 5 readiness pillars — GCP environment, data estate, use case prioritisation, security posture, and team capability — with a self-assessment scorecard and Vertex AI stack selection framework.

GCP	Vertex AI	Gemini	Agent Builder	Enterprise AI

Produced by **Kovil AI Engineering Team** · kovil.ai · June 2026

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01 / Introduction

The Vertex AI Opportunity

Google Cloud Vertex AI is the enterprise platform for building, deploying, and scaling AI agents in production. It brings together 150+ foundation models through Model Garden (including Gemini 2.0 Flash, Gemini 2.0 Pro, Claude 3.5 via Model Garden, and Llama 3.3), Vertex AI Agent Builder for managed agent creation, Reasoning Engine for complex multi-step orchestration, Vertex AI Search for enterprise RAG, BigQuery ML for data-native modelling, and Document AI for structured extraction.

Unlike the Gemini API — designed for prototyping and consumer use cases — Vertex AI adds the enterprise controls organisations require: VPC Service Controls for network isolation, IAM and CMEK for data governance, audit logs for compliance, and Vertex AI Pipelines for MLOps. For any production enterprise deployment, Vertex AI is the correct choice.

> **Who this guide is for:** CTOs, engineering leads, and AI programme managers evaluating whether their organisation is ready to build production AI agents on GCP. Complete the scorecard in Chapter 7 to get your readiness score.

The 5 Readiness Pillars

Kovil AI has identified five pillars that determine whether an organisation can successfully build and sustain production Vertex AI agents. This guide walks through each pillar, provides diagnostic questions, and concludes with a self-assessment scorecard and stack selection framework.

01 **GCP Environment**

Is your GCP project configured with the right APIs, IAM, and networking baseline to host Vertex AI workloads?

02 **Data Estate**

Is your data in GCP, clean, structured, and accessible to AI agents at query time?

03 **Use Case Prioritisation**

Have you identified the highest-ROI AI agent use case to pilot — and do you have a clear success metric?

04 **Security Posture**

Are VPC Service Controls, CMEK, IAM, and audit logging configured to meet your compliance requirements?

05 **Team Capability**

Does your team have the GCP skills, agent architecture knowledge, and MLOps practices to build and sustain this?

02 / Pillar 1

GCP Environment Readiness

Before any Vertex AI workload can run in production, your GCP environment must be correctly configured. Environment readiness is the most common bottleneck we encounter — organisations underestimate the configuration work required to go from "we have a GCP account" to "we can run a production AI agent."

What environment readiness covers

GCP Project Structure

Separate projects for dev, staging, and production. Vertex AI workloads should not share a project with other GCP services in production — this affects billing isolation, IAM scope, and quota management.

API Enablement

Vertex AI API, Vertex AI Agent Builder API, Vertex AI Search API, BigQuery API, Cloud Storage API, Secret Manager API, and Cloud Logging API must all be enabled. Missing API enablement is the single most common early blocker.

IAM Configuration

Service accounts for Vertex AI workloads need the correct minimum-privilege roles: Vertex AI User, BigQuery Data Viewer, Storage Object Viewer, and Secret Accessor. Over-permissioning is a security risk; under-permissioning causes runtime failures.

Networking Baseline

VPC configuration for Vertex AI workloads — including Private Google Access for Cloud NAT, firewall rules for Vertex AI endpoint access, and (for regulated industries) VPC Service Controls perimeter setup.

Quota Configuration

Vertex AI has per-region quotas on model requests per minute and concurrent prediction requests. Production workloads require quota increase requests submitted 1–2 weeks before go-live.

Region Selection

Choose your primary region before any deployment. Vertex AI has regional model availability variations — not all models are available in all regions. Data residency requirements (GDPR, HIPAA) must be satisfied at the region level.

> **Kovil AI environment audit:** Our Strategy & Readiness engagement includes a full GCP environment audit — reviewing project structure, IAM, networking, and quota against the Vertex AI production checklist. Most organisations have 8–15 configuration gaps before their first agent workload can run.

03 / Pillar 2

Data Estate Assessment

AI agents are only as good as the data they reason over. Data estate readiness determines whether your agents will answer accurately and whether they can be grounded in authoritative, up-to-date information. This pillar is where most organisation underestimate the work required.

The data readiness dimensions

Data Location

Is your data already in GCP? Vertex AI Search can index data from Cloud Storage, BigQuery, Cloud SQL, and website content. Data that lives in on-premise systems, AWS S3, or Azure Blob Storage requires migration or connector setup before agents can access it.

Data Quality

Agent accuracy is directly correlated with data quality. Incomplete records, inconsistent field formats, duplicated entries, and stale data degrade agent performance in ways that are difficult to debug post-deployment. A data quality audit before agent build is strongly recommended.

Data Structure

Structured data (BigQuery tables with clear schemas) is the easiest to ground agents in. Semi-structured (JSON, XML in GCS) requires parsing configuration. Unstructured (PDFs, Word docs, email) requires chunking and embedding strategy — which Vertex AI Search handles, but with configuration choices that significantly affect retrieval quality.

Data Freshness

How frequently does your data change, and how quickly must agents reflect those changes? Vertex AI Search supports scheduled re-indexing and real-time updates via the data ingestion API. Your freshness requirements determine the indexing architecture.

Data Access Control

In regulated industries, agents must only surface information the querying user is authorised to see. Vertex AI Search supports access control lists (ACLs) that map to IAM identities — but this must be configured deliberately. An agent without access control will happily surface restricted data to any user.

BigQuery as the data backbone

Organisations with data already in BigQuery are in the strongest position for Vertex AI agent builds. BigQuery ML allows ML models to run directly against the data warehouse — no data movement, no ETL, no latency from API calls to an external ML service. Gemini in BigQuery (BQML integration) allows SQL queries to invoke Gemini for summarisation, classification, and entity extraction inline with analytics queries.

04 / Pillar 3

Use Case Prioritisation

The most common mistake in enterprise AI programmes is starting with the most technically interesting use case rather than the highest-ROI use case. Vertex AI agents are expensive to build correctly — scoping the first use case well is the difference between a programme that generates demonstrable value in three months and one that produces a proof of concept that never reaches production.

The use case selection framework

Dimension	High Score (3)	Medium Score (2)	Low Score (1)
Data readiness	Data in GCS/BQ, clean, updated daily	Data in GCP, needs cleaning	Data outside GCP or poor quality
Query volume	>5,000 queries/month on this use case	500–5,000/month	<500/month
Resolution clarity	Clear right/wrong answer (FAQ, lookup)	Semi-structured decision	Open-ended judgement call
Labour cost	High-cost manual process (>$200k/year)	Moderate cost ($50k–$200k)	Low-cost process (<$50k)
Risk tolerance	Low-risk, reversible actions	Medium risk with human review	High-risk or irreversible actions
Stakeholder buy-in	Executive sponsor identified	Department-level support	No clear owner

Score each candidate use case across these six dimensions. Use cases scoring 14+ are strong candidates for your first pilot. Use cases scoring under 10 should be deferred until data and infrastructure are ready.

05 / Pillar 4

Security Posture

Enterprise AI on GCP requires a deliberate security configuration. The good news: Google Cloud Vertex AI provides more enterprise security controls than any other AI platform — VPC Service Controls, IAM, CMEK, audit logging, and data residency guarantees. The risk: those controls only protect you if they are configured correctly.

VPC Service Controls

VPC Service Controls create a security perimeter around your Vertex AI workloads, preventing data exfiltration by restricting which services can communicate with Vertex AI APIs. This is a hard requirement for HIPAA, PCI-DSS, and SOC 2 Type II deployments. Configuration is non-trivial — incorrect perimeter rules break agent functionality in ways that are difficult to debug.

Customer-Managed Encryption Keys (CMEK)

By default, Google encrypts Vertex AI data at rest with Google-managed keys. CMEK allows you to use your own Cloud KMS keys — a requirement for many regulated industries and a strong preference for organisations with sensitive IP in their training data or agent prompts.

IAM Least Privilege

Vertex AI service accounts should follow least-privilege: only the specific roles required for the workload, scoped to the specific project. The common mistake is using owner-level service accounts during development and carrying that into production. We audit IAM configuration as part of every pre-launch review.

Audit Logging

Cloud Audit Logs for Vertex AI captures all API calls — model invocations, data access, configuration changes. For compliance, audit logs must be enabled, retained for the required period, and exported to a SIEM. This is often configured after the fact — it should be configured before the first production request.

Data Residency

Vertex AI allows you to specify the region where data is processed and stored. GDPR Article 44 requirements and data sovereignty laws (UK, EU, India, Australia) must be satisfied at the region level. Choosing the wrong region at build time means a migration later.

Einstein Trust Layer equivalent

Vertex AI does not have a branded safety layer like Salesforce, but provides equivalent controls: Responsible AI safety filters, custom model safety thresholds, output monitoring, and the ability to run PII detection (via DLP API) on all agent inputs and outputs before logging.

06 / Pillar 5

Team Capability

Building production Vertex AI agents requires a specific combination of skills that most organisations do not have in-house — and that is not a criticism. The Vertex AI stack is 18 months old in its current form; there is no large pool of engineers with 3+ years of Vertex AI Agent Builder and Reasoning Engine experience. Understanding your team gaps honestly is the fastest path to a successful build.

Skills required for production Vertex AI agents

GCP Platform Engineering VPC, IAM, Cloud KMS, Secret Manager, Cloud Logging, Cloud Run, GKE — the GCP platform skills that form the foundation every Vertex AI deployment runs on.	**Required**
Vertex AI SDK & APIs Python SDK for Vertex AI, Agent Builder API, Reasoning Engine SDK, Vertex AI Search API, BigQuery ML SQL extensions.	**Required**
LLM Prompt Engineering System prompt design, few-shot examples, grounding instruction, persona definition, safety instruction — the engineering of agent behaviour through prompts.	**Required**
Agent Orchestration Reasoning Engine deployment, LangChain / LlamaIndex integration, multi-step tool calling, memory management, session handling.	**Required for complex agents**
BigQuery ML SQL-based ML model creation, BQML model types (logistic regression, boosted trees, time series), Gemini in BigQuery integration.	**Required for data-intensive agents**
MLOps Vertex AI Pipelines, Model Monitoring, Explainable AI, automated retraining triggers, canary deployment.	**Required for sustainable operations**
RAG Architecture Chunking strategy, embedding model selection, hybrid retrieval tuning, re-ranking, hallucination guardrails, retrieval evaluation.	**Required for knowledge-grounded agents**

07 / Self-Assessment

Readiness Scorecard

Complete this scorecard honestly before your Vertex AI programme kickoff. For each statement, tick: ✓ Fully in place / ~ Partially in place / ✗ Not yet.

Statement	✓	~	✗
GCP Environment			
We have separate GCP projects for dev, staging, and production.	■	■	■
All required Vertex AI APIs are enabled in our production project.	■	■	■
IAM service accounts for AI workloads follow least-privilege principles.	■	■	■
Our VPC is configured with Private Google Access enabled.	■	■	■
We have submitted quota increase requests for Vertex AI endpoints.	■	■	■
Our primary region is selected and satisfies data residency requirements.	■	■	■
Data Estate			
The data our agents will reason over is stored in GCS or BigQuery.	■	■	■
We have run a data quality audit on the target data sources.	■	■	■
We understand the freshness requirements and have an indexing schedule.	■	■	■
Access control lists are in place for sensitive data in Vertex AI Search.	■	■	■
We have a data governance policy that covers AI agent data access.	■	■	■
Use Case			
We have identified and agreed our highest-ROI pilot use case.	■	■	■
We have defined a measurable success metric for the pilot.	■	■	■
The use case involves data that is ready for agent access today.	■	■	■
We have executive-level sponsorship for the AI programme.	■	■	■
We have completed the use case scoring framework in Chapter 4.	■	■	■
Security			
VPC Service Controls perimeter is configured for Vertex AI.	■	■	■
CMEK is enabled or evaluated for our data classification requirements.	■	■	■
Cloud Audit Logs for Vertex AI are enabled and exported to SIEM.	■	■	■
We have completed a threat model for our AI agent architecture.	■	■	■
Team			
At least one team member has GCP Professional ML Engineer certification.	■	■	■
We have Python SDK experience on the team (or a partner engaged).	■	■	■

We have a plan for ongoing agent monitoring and optimisation post-launch.	■	■	■
We have engaged or evaluated a Vertex AI implementation partner.	■	■	■

Scoring: Count your ✓ marks. 18–24: Ready to build. 10–17: Address gaps before building. Under 10: Engage a partner before starting.

08 / Framework

Vertex AI Stack Selection Framework

The right Vertex AI stack depends on your use case, data structure, team capability, and latency requirements. This framework maps the most common agent types to the recommended Vertex AI components.

Use Case Type	Recommended Stack	Avoid
Customer-facing FAQ / service deflection	Agent Builder + Vertex AI Search (RAG) + Gemini Flash	Reasoning Engine (over-engineered for this pattern)
Complex multi-step internal agent (e.g. SDR research, ops automation)	Reasoning Engine + LangChain/LlamaIndex + Gemini Pro	Agent Builder (insufficient orchestration control)
Structured data analytics / prediction	BigQuery ML + Gemini in BigQuery + Looker	Vertex AI Search (wrong tool for structured data)
Document processing & extraction	Document AI + Vertex AI Search + Gemini Vision	BigQuery ML (wrong tool for unstructured docs)
Real-time personalisation	Vertex AI Matching Engine + Bigtable + Gemini embeddings	Vertex AI Search (latency too high for real-time rec)
MLOps / model lifecycle management	Vertex AI Pipelines + Model Registry + Model Monitoring	Custom orchestration (re-inventing managed infrastructure)

09 / Next Steps

Working with Kovil AI

Kovil AI is a specialist AI engineering firm that builds production Vertex AI agents for enterprise organisations. Every engagement is outcome-focused — we work alongside your team from discovery through to a working agent in production.

How we work with you

01

Discovery Call

A free 45-minute call with a Kovil AI engineer. We learn about your organisation, data landscape, and the use cases you have in mind. No sales pressure — this is a technical conversation.

02

Scoping & Architecture

We review your GCP environment, data estate, and candidate use cases. Working with your team, we define the agent architecture, integration points, data requirements, and what "done" looks like — including measurable success criteria.

03

Statement of Work

A clear, written SOW covering scope, deliverables, milestones, timeline, and acceptance criteria. Fixed scope, no ambiguity. You know exactly what you are getting before any work begins.

04

Milestone-Based Delivery

We build in sprints tied to agreed milestones — environment setup, first agent deployed to staging, evaluation results reviewed, production deployment. You review and sign off at each milestone before we proceed.

05

Project Completion & Handover

Full documentation, runbooks, and a handover session with your engineering team. We do not disappear at go-live — post-launch support is included in every engagement so your team can maintain and evolve the agent with confidence.

> **Start with a Discovery Call:** 45 minutes, free, no obligation. A Kovil AI engineer will review your Vertex AI readiness scorecard, identify the top gaps, and outline what your first production agent could look like. kovil.ai · info@kovil.ai · Book at kovil.ai/engage/outcome-based-project